

# CALCULUS NOTES

NELLY

## 1. INTRODUCTION TO CALCULUS

The concept of a limit formalizes approaching behavior. As shown in [1].

**Theorem 1.1.** *If  $f(x) = x^2$ , then*

$$\lim_{\{x \rightarrow a\}} f(x) = a^2.$$

*Proof.* Polynomials are continuous. □

## 2. DERIVATIVES

**Theorem 2.1.** *The Riemann hypothesis is true.*

*Proof.* This is left as an exercise to the reader, given the complexity of the theorem. □

## REFERENCES

1. Strang, G.: Calculus. Wellesley-Cambridge Press (2016)